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Vectors, Parametric Equations and Three Dimensional Coordinates

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Contents

To the Instructor

This module is a 2-3 week introduction to vectors, parametric equations, and three-dimensional space. We have primarily used it at the Naval Academy in Calculus II, towards the end of the course. We have also used it in a Calculus III course for first year students, at the very beginning. Most of our sections have 18-20 students, and meet for four 50-minute class sessions per week; in this setting we have found that the unit takes about 9-10 class sessions. It is broken up into 9 sections, corresponding to the homework for each of 9 days. (For source files or a differently formatted version, contact the author.)

In the Calculus II class, this unit comes after series and integration, and immediately after 1-2 days on polar coordinates. For series and integration, we use Mairead Greene and Paula Shorter's Calculus II problem sequence (from iblcalculus.com). During that part of the course, students encounter material for the first time during class, working in small groups. When we start the vectors unit you are reading now, we change the format so that students' first encounter with the material is outside class, and class time is focused on individual student presentations of the work. We have noticed that the students' prior experience with group work increased their independence, which made this unit more successful.

Calculus courses at the Naval Academy have a common textbook (Stewart's *Calculus: Early Transcendentals*), and a course coordinator who writes a common final exam and a suggested course calendar. For the past two years we have had roughly half of the Calculus II instructors using inquiry-based learning (including this unit), and the other half following the textbook more closely. The students in both groups do about equally well on the common final exam.

To be successful in this unit, students should already be able to do the following.

- Plot points in the plane and find the equation of a line in the plane.
- Know that velocity is a derivative of position (or displacement), and know the difference between velocity and speed. (These ideas will be reinforced, and made more precise.)
- Know the "Pythagorean" theorem and be able to apply it to find lengths

of sides of a right triangle.

- Know or be able to compute values of sin and cos for standard angles.

There are a few places where it is helpful for students to know that force can be measured in Newtons ($1\text{N} = 1\text{kg m/s}^2$), and that "work equals force times distance" and can be measured in joules ($1\text{J} = 1\text{N m}$). We find that a critical mass of our students know these things from high school physics. There is at least one place where it is helpful for students to know what "port" and "starboard" mean.

Learning Objectives

By the end of this module, students should expect to be able to do the following.

- Use three-dimensional coordinates to represent locations in a marked three-dimensional space.
- Give examples of quantities that can be well modeled using vectors.
- Translate between algebraic and geometric representations of a vector.
- Explain the operations of vector addition, scalar multiplication, dot product and cross product in both algebraic and geometric representations.
- Describe plane curves and 3D lines using parametric equations.
- Explain, using the dot product, the formula for the equation of a plane through a given point perpendicular to a given vector.

The material covered corresponds to Sections 12.1-12.5 of Stewart's *Calculus: Early Transcendentals, 8e*.

Most of these topics are new to most of our students. Students who have already taken a calculus course in high school which covers these topics still benefit greatly by deepening their understanding. The students who already know how to compute the cross product try to teach their method to their classmates, and inevitably it is correct, but different from the one presented. (I think I learn a new way to calculate cross product every time I teach this course!) I used to discourage this, but now encourage it, because the resulting confusion and sign errors lead to good discussions and more robust understanding.

Daily Implementation

Each day at the end of class, we hand out the batch of problems for students to work on before the next class. At the beginning of the next class, most of us take volunteers to present each problem; some do this via an online signup before class. Some of us assign problems to students randomly; creating a deck of cards with student names and/or photos on it can be a useful tool! Some of us assign two or three students to each problem, so that everyone is presenting something every day. Some of us assign one student to each problem, but assign a group to a problem that the whole class was stuck on. The presenters write their solutions on the board all at once. If there are students not at the board, they can be checking what's going up against their own work, or thinking about any problems they were stuck on. The instructor monitors what's going on the board, asks questions and makes suggestions. The purpose of the questions and suggestions is not to make sure that the correct answer goes up, but to make sure that the thoughts are communicated clearly. In fact, it is often better to have a common wrong answer on the board, so that it comes up in discussion.

One the solutions are up, then we go around the room discussing each problem. This is sometimes done with the student at the board, but sometimes the instructor will be the one at the board annotating the student work. The goal of the discussion is for the class as a whole to make sense of what is going on mathematically; recording "the correct answer" (or one or more correct answers, in many cases!) should be a byproduct of this understanding, not the primary goal. In this module I have included notes after almost every problem, indicating what I think is most important to draw out in the discussion, how the problems fit together, etc. I've also included, for quick reference and reassurance, answers to the computational problems. I try to never give a student the answer, asking instead for them to check it mathematically (if they have the tools), whether it makes sense (e.g. did we get a negative number for area?), or to confirm with classmates. As a last resort to save time I will say, "That's what I got, too."

The problems for each day will have three or four themes, each part of a thread that will continue to the next day or is continued from previous days. The amount of material per day is fairly well calibrated by now, but there are still occasional times when the discussion takes a different amount of time than I expect. If you have extra time, students can always start working on the next day's problems during class. In case you are running out of time at the end of class, I have tried to make the last few problems ones that are either easy computations that can be checked with little to no discussion, or problems that can be put off until the next day.

Assessment and Grades

Among the instructors who use this problem sequence, we have used several different grading schemes. Some instructors give credit for student presentations, or for homework. If homework is being collected for a grade the day the problems are presented, the grade should either be based on how many were attempted, or what strategies were tried, or some other measure of effort. It is important that students know that they are not expected to get correct solutions to all of the problems before class! All of us give quizzes, including questions about computations (e.g. "Find parametric equations for the line through [two given points]") and basic proofs (e.g. "Prove that if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in three dimensions, then $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w}) = (\mathbf{u}\cdot\mathbf{v})+(\mathbf{u}\cdot\mathbf{w})$.") Some of us have used standards-based grading, allowing students to retake quizzes multiple times. For more details on the standards-based grading scheme we have used, see the forthcoming article in *PRIMUS*, "Raising the Bar with Standards-Based Grading," by the author, Megan Selbach-Allen, Sarah J. Greenwald, and Jill Thomley.

Throughout the semester we give four midterm tests; the material in this unit typically appears all on one 50 minute test, with some problems from other units. The common final exam consists of 20 multiple choice questions and 10 long-answer questions; sample old final exams for both Calculus II and Calculus III (SM221P) can be found on the U,. S. Naval Academy math department website.

A typical grade breakdown might be:

- Final exam 30%
- Midterm tests 40%
- Daily work (presentations and/or homework) 10%
- Quizzes and writing assignments 20%

Mathematical Quirks

There are many places where standard textbooks (including Stewart) will use the position vector of a point, where I have instead chosen to use the point itself. This means that sometimes we subtract one point from another to get a vector, or add a vector to a point to get another point. See for example the parametric equation immediately following Problem 24. We do not add points to each other.

Philosophically, there are two reasons I know of to use the position vector. Mathematically, it is comforting because addition is then taking place in a vector space. Physically, it helps to reinforce the idea of "displacement" and remind us that all positions are relative to a reference point.

On the other hand, when students are doing the geometric reasoning to understand equations of lines and planes, I find it is helpful to keep the distinction between points and vectors clear. Physically, the idea of position is more useful in this context than the idea of displacement. Mathematically, under the hood, we are thinking of the set of vectors as a vector space (and in particular, a group under addition) and the set of points as a manifold. Adding two vectors together is happening in the vector space; adding two points to each other doesn't make sense (as, for most manifolds, it does not). Adding a vector to a point defines an action of the group on the manifold. When we subtract one point from another, we are asking for the group element that takes one point to the other (using the facts that the action is transitive, and that this element is unique). This is a more elaborate mathematical setup than just putting everything on the same footing in a vector space, but I believe it does a much better job of capturing and supporting student thinking.

To the Student

In your mathematical career, you have been able to make good use of the formula $y = mx + b$ for a line in the plane. The main goal of the next two weeks will be to expand your repertoire of formulas for lines and planes in three dimensional space.

One of the main tools we will use is the idea of a "vector." A vector is designed to represent quantities with both size and direction. Vectors are used to represent velocities and forces in many different applications: wind speeds, ocean currents, the forces acting on a bridge span, the motions of subatomic particles and galaxies. Vectors can also represent prices on the stock market, changes in political opinions, DNA sequences, and much much more. The operations on vectors that you learn in this module will have meaning in each of these different contexts. We hope that this short unit will give you a taste for these possibilities, and that you will seek to learn more.

Each day there will be a batch of problems for you to work on and think about before class. If you read a problem and don't know how to do it, read it again more slowly. Sometimes drawing a picture of what it says can help. If you've thought about it for five minutes and are still stuck, go on to the next problem and come back to it later. During class you will have an opportunity to ask questions and find out the answer.

Chapter 1

Problem Sequence

1.1 Day 1

Work on the following problems outside of class, on your own. Bring your ideas and solutions to class to discuss.

Problem 1. *Let $f(t) = (t-1, 3-t)$. Sketch the points $f(0)$, $f(1)$, $f(2)$, $f(3)$, and $f(4)$ in the plane. What do you notice?*

It's helpful if the student presenting this also writes the coordinates of the five points on the board. The students always answer: they're all on a line! Followup question: What line is it? (Let the students figure out: $y = 2 - x$.) Mini-lecture: We call $f(t)$ a **parametric equation** for this line. The "parameter" is t. Is $y = 2 - x$ true for all values of t? (Give the students a minute to check the points; give space for someone to suggest plugging $y = 3 - t$ and $x = t - 1$ into $y = 2 - x$, but show them this even if no one suggests it.) This is sometimes called "eliminating the parameter."

Problem 2. *Suppose that $f(t)$ in the previous problem represents the position of a ferret at time t.*

(a) In what direction is the ferret traveling?

(b) What is the speed of the ferret?

Almost always, the students have internalized that "slope = speed." But in this case, the slope is not the speed, because time is not even on the graph!

Questions to fuel the discussion:

- The slope of the line is -1. Is the speed 1? Is the ferret walking backwards, or downhill?
- Suppose time is measured in hours from noon. Where is the ferret at noon? Where is the ferret at 2pm? How far did it travel?

- Where is the t axis on this graph?

Answers: Direction: Southeast. Speed $\sqrt{2}$ mph or m/s or whatever units we're using.

Mini-lecture: We can use a vector to represent the velocity of the ferret. Geometrically a vector is an arrow with direction and size (represented by length); algebraically this particular vector is $\langle 1, -1 \rangle$.

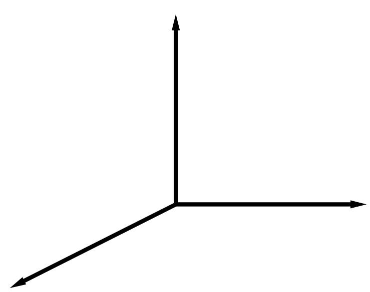
Problem 3. *Imagine you have a young cousin who doesn't yet know about (x,y) coordinates in the plane. How would you explain it to them? As an example, how do you describe plotting the point* $(-2, 3)$*?*

The point of this problem is to "activate prior knowledge", and have them clearly articulate a familiar thing so that they can more easily modify it for a new situation.

I learned this as "First you go over to the ladder, then you climb up (or down)."

Problem 4. *We would like to introduce a similar way of locating points in three dimensions. We use three axes, which we call the x, y, and z axes. In mathematics we typically use the convention that the positive z axis points up, the positive y axis points to the right, and the positive x axis points out of the page towards us.*

Label the axes in the picture below. (Imagine looking into the corner of a room.)



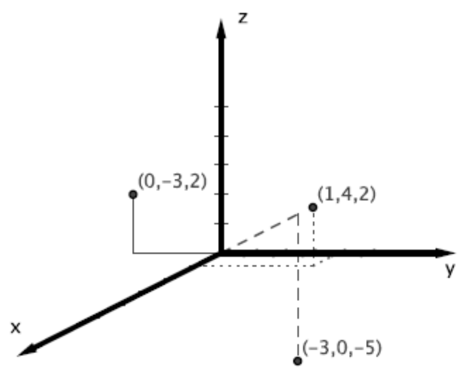
This turns out to be an exercise in reading carefully! This might be a good time to point out that throughout this unit, there will be problems that refer directly to information in the paragraph right before the problem.

Make sure that everyone can see how this diagram represents three dimensions. Pointing to a corner of the room and having a whole-class discussion of where the x, y, and z axes would be relative to that corner is helpful. Some students might benefit from actually holding their paper up and moving so that it looks like the corner of the room.

I also mention to them that in their physics and engineering classes, they are likely to see another convention where the z axis is the one coming out of the page / board, and the x and y axes look like they do in 2D.

Problem 5. *Plot the following points on the three-dimensional coordinate system above:* $(1,4,2)$, $(-3,0,-5)$, *and* $(0,-3,2)$.

Often the tricky part for the students is how to represent motion that changes the x coordinate only. For example, the point at $(1,4,0)$ shouldn't look in the picture like it's straight down from the point $(0,4,0)$.



(Answers)

Problem 6. *Draw a new 3D coordinate system, label the axes, and sketch these planes.*

(a) $z = 5$

(b) $x = 2$

(c) $y = -3$

In the discussion of this problem, it can be fun to use the corner of the room as the origin, meters as the units, and ask students to stand on the $x = 2$ plane. Then explain where the $y = -3$ plane is and why they can't stand there (it's in another classroom or outside maybe). Then stand on the $z = 1$ plane, (i.e. on their desks!). Not just for fun, but because it helps the students embody the knowledge.

It's also worth talking about why these shapes form planes and not lines. I've been surprised to find that students are not always clear on the difference!

Problem 7. *In two dimensions, two lines must either intersect or be parallel. Is the same thing true in three dimensions? What about two planes, or a line and a plane? Sketch and/or describe the different possibilities.*

Sometimes students think ”parallel” *means* non-intersecting. But we want parallel to mean really going the same direction. This is a good time to introduce the term “skew lines”.

In your mathematical career, you have been able to make good use of the formula $y = mx + b$ for a line in the plane. The main goal of the next two weeks will be to expand your repertoire of formulas for lines and planes in three dimensional space.

Note to the instructor: There is time here to add two or three review problems from previous material.

1.2 Day 2

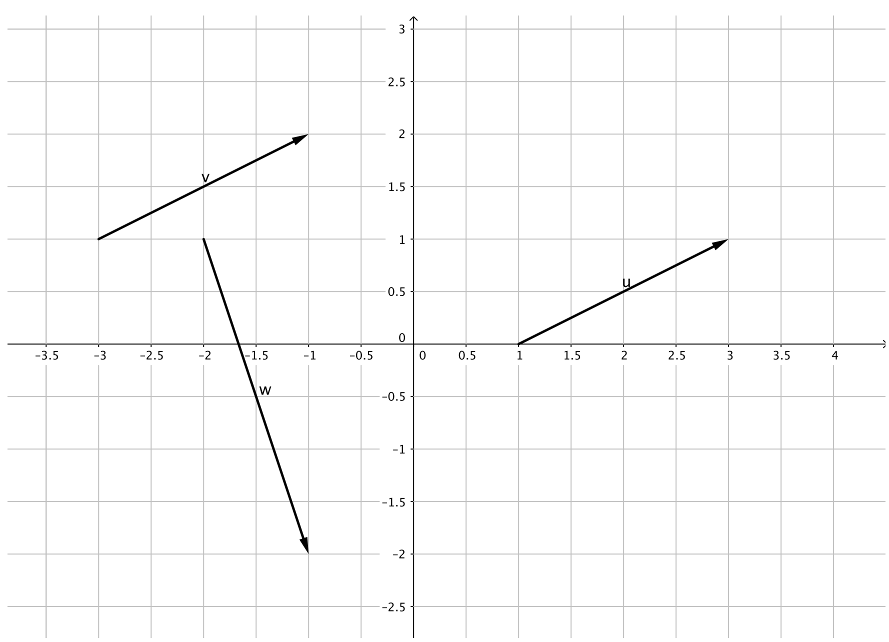
Definition 1. *A* ***vector*** *is an arrow with a direction and a length ("magnitude") but no fixed position. We can represent a vector in the plane as an ordered pair* $\langle a,b\rangle$.

Problem 8. *The velocity of the ferret from Problem 2 is (fill in the blanks)* $\langle\quad,\quad\rangle$.

This might be a good one to do as a whole class discussion without having a student present it. Whoever presents it should reproduce the sketch and equations from Problems 1 and 2.

Answer: $\langle 1,-1\rangle$. We could say that every hour, the ferret travels both one unit east and one unit south.

Note that here we are not giving a formal definition for the notation $\langle a,b\rangle$, but appealing to students' intuition about what the definition should be, i.e. what it should represent. This reflects mathematical practice– in research mathematics, we often start using notation for things, then later formally define the notation.



In the picture above, vectors **u** and **v** are both $\langle 2,1\rangle$. Vector **w** is $\langle 1,-3\rangle$.

Notice that here and in the textbook, names of vectors are bold letters like $\mathbf{v}$. On the blackboard, we usually write $\overrightarrow{v}$ or $\bar{v}$ or $\underline{v}$ to indicate a vector.

Problem 9. *The vector from* $(1,3)$ *to* $(5,-2)$ *is (fill in the blanks)* $\langle\quad,\quad\rangle$. *Illustrate with a sketch.*

$\langle 4,-5\rangle$.

Problem 10. *Why are vectors* $\mathbf{u}$ *and* $\mathbf{v}$ *in the picture above both* $\langle 2,1\rangle$*?*

(Thanks to the referee for suggesting this as a problem!) The reasoning should be because of what the vectors are doing, not because "It says right there..."

Problem 11. *The vector from* (a,b) *to* (c,d) *is (fill in the blanks)* $\langle\qquad,\qquad\rangle$.

Answer: $\langle c-a, d-b\rangle$.
This is now a definition!

Problem 12. *The vector from* $(0,0)$ *to* (a,b) *is (fill in the blanks)* $\langle\qquad,\qquad\rangle$. *This is sometimes called the "position vector" of the point (a,b). Draw a sketch of the point* $(1,5)$ *and its position vector.*

Answer : $\langle a,b\rangle$. Make sure that the distinction between the point and the vector is clear in the sketch, and the distinction between $\langle,\rangle$ and $(,)$ is clear as well.

Problem 13. *Where does the plane* $2x+3y+z=6$ *intersect the three coordinate axes? Use this information to sketch the plane.*

Hints to give, if a student is stuck: If you're on the x axis, what is your y value? What is your z value?

Answer: $(3,0,0)$, $(0,2,0)$, and $(0,0,6)$. Standard practice is to plot these three points, then draw the triangle between them, and call that a sketch of the plane. Students often struggle with representing an infinite plane by a triangle, so it's worth discussing the fact that the plane is infinite, and how it continues beyond the triangle. Using the corner of the room is useful in this discussion, and holding up a sheet of paper with one or more pencils representing coordinate axes is too.

Problem 14. *Sketch and describe the set of points in 3D where* $x=2$ *and* $y=-3$.

Answer: A vertical line through $(2,-3,0)$. Some students may read this as two separate problems, each asking for a plane, but the intent is to identify the intersection of the two planes.

An important followup question: What is "the equation" of this line? Is there just one equation for this line, or do we need both? (I have a smart aleck colleague who would point out here that $(x-2)^2+(y+3)^2=0$ is one equation for this line, so I try to say "linear equation" or "degree 1 equation" when I think it's not too distracting. If a student seems bored you could challenge them to find one degree 2 equation.)

Problem 15. *Sketch the triangle with corners* $(0,0,0)$*,* $(1,3,0)$*, and* $(1,3,2)$*. Explain why this is a right triangle, and use the Pythagorean Theorem to find the distance from* $(0,0,0)$ *to* $(1,3,2)$*.*

Answers: It probably doesn't look like a right triangle, but it is, because one leg of the triangle is perpendicular to the xy plane, and the other leg is in the xy plane. To get the distance, use the Pythagorean Theorem twice: Once to get the distance from $(0,0,0)$ to $(1,3,0)$, then again to get the distance as $\sqrt{1^2+3^2+2^2}$. In the discussion after this problem, point out the structure as $\sqrt{a^2+b^2+c^2}$.

Problem 16. *Let* $q(t)=(2\cos t, 2\sin t)$*. Use graph paper to plot the points* $q(0)$*,* $q(\pi/4)$*,* $q(\pi/2)$*,* $q(2\pi/3)$*,* $q(5\pi/6)$*,* $q(\pi)$*,* $q(3\pi/2)$*, and* $q(2\pi)$*. What do you notice?*

If you are running out of time in class, put off discussion of this problem to the next day and make sure to discuss Problem 17.

It's a circle! Reinforcing polar coordinates with $x=r\cos\theta$ and $y=r\sin\theta$. Contrary to how the book does it, there are no vectors in this problem. The output of q is a point in the plane.

Problem 17. *Suppose that a giraffe is traveling along the same path as the ferret from Problem 2, but going twice as fast in the opposite direction. Write an equation or set of parametric equations for the position of the giraffe at time t.*

I'm not specifying where the giraffe starts, so there are many correct answers. One is $g(t)=(2-2t,2t)$, or $x=2-2t$ and $y=2t$. In class it's worth plotting to check the path is the same– students are likely to double the position as well as the velocity.

This problem is important because we will use "twice as fast in the opposite direction" to motivate scalar multiplication, first thing tomorrow!

1.3 Day 3

Consider the velocity of the ferret from Problem 2. If we call that $\mathbf{v}$, then the velocity of the giraffe from Problem 17 will be $-2\mathbf{v}$. This is an example of multiplying a vector by a number.

Problem 18. *Just to make sure we're all together on this. The velocity of the ferret was*

$\mathbf{v} = \langle \quad , \quad \rangle$

and the velocity of the giraffe was

$-2\mathbf{v} = \langle \quad , \quad \rangle$.

Again it's a good idea to get the graph from Problem 17 on the board, and use this to review some ideas from the previous day.

When we multiply a vector by a number, the number acts by scaling the size of the vector, and if the number is negative, by changing the direction. Because of this scaling effect, numbers in this context are often called *scalars*. When things get complicated, it is often very useful to keep track of which of the things you're talking about are vectors and which are scalars. Multiplying a vector by a scalar is called *scalar multiplication.*

Problem 19. *If* $\mathbf{u} = \langle 3,5 \rangle$*, then* $4\mathbf{u} = \langle \quad , \quad \rangle$.

Answer: $\langle 12,20 \rangle$

Problem 20. *If* $\mathbf{v} = \langle a,b \rangle$*, then* $k\mathbf{v} = \langle \quad , \quad \rangle$.

Answer: $\langle ka,kb \rangle$

Problem 21. *If* $\mathbf{w} = \langle a,b,c \rangle$*, then* $k\mathbf{w} = \langle \quad , \quad , \quad \rangle$.

$\langle ka,kb,kc \rangle$. This is another place where we are not starting with the formal definition, but coming up with it based on intuition.

Problem 22. *If* $\mathbf{v} = \langle -2,1,7 \rangle$*, then* $-5\mathbf{v} = \langle \quad , \quad , \quad \rangle$.

$\langle 10,-5,-35 \rangle$.

Problem 23. *Sketch and describe the shape with parametric equations* $x = -2+t$, $y = 1$, *and* $z = -3+t$.

A line in the plane $y = 1$. Follow up discussion: Can we "eliminate the parameter" to get an equation / equations for the line? If you need to cover "symmetric form" for a line, this is a good place to bring it in.

Problem 24. *Does the equation* $z = x-1$ *describe a line or a plane in three dimensions?*

A plane, because y could be anything. So if we want to describe a plane we can use one equation, but if we want to describe a line using linear equations we will need two of them.

This is a good time to go back to thinking of a corner of the room as the origin, and describing (or having students describe) where the plane is, relative to the room.

You may also want to prepare in advance a 3D graph (using Geogebra or Maple or Mathematica or Sage or DPGraph or ...) showing this plane relative to the axes, and the line from Problem 23 in it, viewable from different angles.

We can write the three equations from Problem 23, as one equation:

$$(x,y,z) = (-2,1,-3) + t\langle 1,0,1\rangle.$$

This is sometimes called the "vector form" of the equation.

Problem 25. *Label the point $(-2,1,-3)$ on your graph for Problem 23. Sketch the vector $\langle 1,0,1\rangle$, starting from that point.*

Follow up discussion: Why did we just do that? Often we know geometric information about a line, and want to write down the equation for the line. How does the geometric information correspond to the equation, in this case?

Problem 26. *Sketch and describe the set of points in 3D where $x+y+z=1$.*

A plane through $(0,0,1)$, $(0,1,0)$, and $(1,0,0)$.

Problem 27. *(a) What is the distance from $(-1,-2,4)$ to $(2,3,-1)$?*

(b) What is the distance from (a,b,c) to (p,q,r)?

Answers: $\sqrt{3^2+5^2+5^2} = \sqrt{59}$. $\sqrt{(p-a)^2+(q-b)^2+(r-c)^2}$.

Problem 28. *(a) What is the distance from $(0,0,0)$ to (a,b,c)?*

(b) What is the length of the vector $\langle a,b,c\rangle$?

Answer: $\sqrt{a^2+b^2+c^2}$, for both.

We use the notation $|\mathbf{v}|$ for the length (or "magnitude") of a vector.

Problem 29. *What is $|\langle 2,-5,1\rangle|$? What is $|\langle -3,-4\rangle|$?*

Answer: $\sqrt{30}$, and 5. Followup question: Is $|\langle 2,-5,1\rangle| = \langle 2,5,1\rangle$? Because the notation looks a lot like absolute value. (No, the magnitude is a number not a vector.)

What if a vector had only one entry? What would $|\langle -4\rangle|$ be?

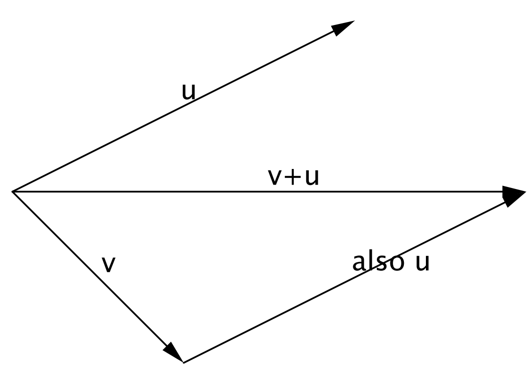
1.4 Day 4

Problem 30. *An aircraft carrier is traveling NW at 5 mph. Meanwhile a sailor is walking briskly across the deck from port to starboard, coincidentally also at 5 mph. What is the net ("resultant") velocity of the sailor? Explain with a picture.*

This is a good place to avoid "proof by authority," or "proof by a formula I know." We should be able to figure it out without any sophisticated mathematics. A good question to ask would be whether the net motion of the sailor is to the east or to the west? Once students can agree that the east and west components of the motion have to cancel out, then the rest is not so bad.

It's not that "The way to solve this problem is by vector addition, but we expected you to either already know that or look it up or somehow magically guess the formula." The truth is that the reason vector addition was defined in the first place was to encode calculations like this, which make sense in the real world, and which people were doing long before they wrote things in vector form.

We define *vector addition* geometrically as the following: If we want to add two vectors, move them so that one of them (say, "**v**") starts where the other (call it "**w**") ends. Then draw the arrow from the beginning of **w** to the end of **v**: that new arrow is $\mathbf{w}+\mathbf{v}$.



Problem 31. *(a) On graph paper, drawing carefully with a ruler, draw two separate arrows and label them* **v** *and* **w**. *You can make them whatever vectors you want.*

(b) Now draw another copy of $\mathbf{v}$ *starting at the end of* $\mathbf{w}$. *(Make sure you don't change the length or direction!)*

(c) Now draw $\mathbf{w}+\mathbf{v}$.

(d) Based on your drawing: What is $|\mathbf{w}+\mathbf{v}|$*?*

If you teach in a room with a document camera, that's a good way for a student to present this one. In this problem and the next one, the act of physically drawing the vectors helps students to "embody" the knowledge of what vector addition is.

The question about magnitude is there to reinforce what that notation means. The answer will be different for each student, depending on their initial choices for **w** and **v**.

Problem 32. *(a) Add the vectors* $\langle 2,4\rangle$ *and* $\langle 3,-1\rangle$ *by drawing them as in Problem 31.*

(b) The vector you get is $\langle 2,4\rangle+\langle 3,-1\rangle=\langle \quad , \quad \rangle$.

Answer: $\langle 5,3\rangle$. It's important that this answer be tied to the geometry. If students find the answer algebraically by adding coordinates, they should be encouraged to go back to the geometric definition to explain why adding coordinates makes sense (this will be revisited in Problem 42).

Problem 33. *Find a vector in the same direction as* $\langle 1,2,-2\rangle$ *but with length* 15.

Answer: $\langle 1,2,-2\rangle$ has length 3, so we need to multiply it by 5 to get $\langle 5,10,-10\rangle$.

Definition 2. *A* unit *vector is a vector whose length (a.k.a. magnitude) is* 1.

Problem 34. *Find a unit vector in the same direction as* $\langle 1,2,-2\rangle$.

Answer: $\langle \frac{1}{3}, \frac{2}{3}, \frac{-2}{3}\rangle$.

Problem 35. *Find a unit vector in the same direction as* $\langle -2,5,2\rangle$.

Answer: $\langle \frac{-2}{\sqrt{33}}, \frac{5}{\sqrt{33}}, \frac{2}{\sqrt{33}}\rangle$.

Definition 3.

- *The unit vector pointing in the positive* x *direction is called* $\mathbf{i}$.
- *The unit vector pointing in the positive* y *direction is called* $\mathbf{j}$.
- *The unit vector pointing in the positive* z *direction is called* $\mathbf{k}$.

Problem 36. *Sketch* $\mathbf{i}$ *and* $\mathbf{j}$ *in the plane.*

This problem and the next one are asking the students whether they read the previous definition, and were able to make sense of it.

Problem 37. *Sketch* **i**, **j**, *and* **k** *on a three dimensional coordinate system.*

Problem 38. *Sketch* $-2\mathbf{i}$, $5\mathbf{j}$, *and* $-2\mathbf{i}+5\mathbf{j}$ *in the plane.*

These are all based on geometry; we hope they will do $-2\mathbf{i}+5\mathbf{j}$ by drawing as in Problem 31. The students might notice that $-2\mathbf{i}+5\mathbf{j}$ is the same as $\langle -2,5\rangle$. But if that doesn't come up here, that's OK because it will tomorrow.

The following is a theorem you may have learned in your trigonometry or precalculus class.

Theorem 1. ***Law of Cosines*** *Given any triangle with sides of lengths* a, b, *and* c, *and having an angle of measure* α *opposite the side of length* a, *the following equation holds:* $a^2 = b^2 + c^2 - 2\,b\,c\,\cos(\alpha)$.

How to prove the Law of Cosines! This is just for fun, not necessary for the students. Draw the line perpendicular to the c side through the opposite corner of the triangle. It cuts the c side into two pieces, of lengths $x = b\cos\alpha$ and $c-x$. It also divides the triangle into two right triangles; use the Pythagorean Theorem on both. (So cool!)

Problem 39. *Sketch the vectors* $\langle 3,1\rangle$ *and* $\langle 1,5\rangle$. *Use the Law of Cosines to find the angle between them.*

Answer: A little more than 60 degrees. Some students will not have figured out how to do this, but will get another chance (with a 3D one) tomorrow after seeing this example. You might ask them whether there is a better way to find the angle between two vectors. In the plane, one could sketch them and use a protractor. But in 3D, that won't work! And the Law of Cosines is great, but can get tedious. But soon we will learn a much easier way!

Problem 40. *Write parametric equations for the line through the point* $(4,0,-3)$ *in the direction of the vector* $\langle 5,-2,1\rangle$.

Answers: $(4,0,-3)+t\langle 5,-2,1\rangle$, or $(4+5t,-2t,-3+t)$, or $x = 4+5t$, $y=-2t$, $z=-3+t$. It's kind of nice to get all three on the board.

1.5 Day 5

Problem 41. *Sketch the vectors $\langle -1,-2,4\rangle$ and $\langle 2,3,-1\rangle$. Use the Law of Cosines to find the angle between them.*

Answer: About 134 degrees.

Problem 42. *In general, is it true that $\langle a,b\rangle + \langle c,d\rangle = \langle a+c, b+d\rangle$? Why or why not?*

Yes. Since we defined vector addition geometrically by drawing the vectors, the explanation for why should include a picture!

Problem 43. *In general, is it true that $\mathbf{v}+\mathbf{w} = \mathbf{w}+\mathbf{v}$? Why or why not? Illustrate with a picture.*

Also yes. Algebraically, because addition is commutative; geometrically it's the other two sides of the same parallelogram.

Problem 44. *Is it always true that $|\mathbf{v}+\mathbf{w}| = |\mathbf{v}|+|\mathbf{w}|$? If so, explain why; if not, give a counterexample.*

No. They probably drew a counterexample yesterday in Problem 31. This problem is here partly to practice proof vs. counterexample, and especially in the context of what the notation $|\mathbf{v}|$ represents. Potential followup question, if you have a lot of extra time: "Can the left hand side ever be larger than the right hand side?"

Problem 45. *How should vector subtraction be defined?*

Questions to fuel the discussion: How do we want subtraction to be related to addition? How does it work with numbers? How should it be related to scalar multiplication by -1?

Answers: Algebraically, by subtracting in each component. Geometrically, if $\mathbf{w}$ and $\mathbf{v}$ start at the same point, then $\mathbf{w}-\mathbf{v}$ goes from the tip of $\mathbf{v}$ to the tip of $\mathbf{w}$.

Problem 46. *How are the vectors $\langle 3,-7\rangle$ and $3\mathbf{i}-7\mathbf{j}$ related?*

They are the same thing.

Problem 47. *How are the vectors $\langle a,b,c\rangle$ and $a\mathbf{i}+b\mathbf{j}+c\mathbf{k}$ related?*

They are the same thing.

Problem 48. *If you drop a $10g$ magnet near a refrigerator, it will not fall straight down but be pulled towards the fridge as it falls. Suppose that the magnetic force is about 4 Newtons, and that acceleration due to gravity is 9.8 m/s^2.*

(a) Draw a diagram representing the two forces on the magnet.

(b) Add the forces to get the total ("resultant") force on the magnet.

The force due to gravity is .098 N straight down, so the resultant force is $\langle 4, -.098\rangle$. Tomorrow there will be a more complicated problem like this.

Problem 49. *Does the line* $P(t) = (2,5,1) + t\langle 3,-1,2\rangle$ *intersect the plane* $x + 2y - 5z = 10$*? If so where?*

To do this they need to unpack the vector form of the line equation into its three components; this could be a good lesson in writing down both sides of an equation like $x = 2 + 3t$, not leaving off the $x =$ half. Then they plug in and solve for t, and need to realize that they aren't done yet.

Answer: Yes, at $(1, \frac{16}{3}, \frac{1}{3})$ (when $t = -\frac{1}{3}$).

Discussion question, if there is time: What does $t = -\frac{1}{3}$ signify?

Definition 4. *If* $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ *and* $\mathbf{w} = \langle w_1, w_2, w_3\rangle$ *then the* dot product *of* $\mathbf{v}$ *and* $\mathbf{w}$ *is defined by*

$$\mathbf{v}\cdot\mathbf{w} = v_1w_1 + v_2w_2 + v_3w_3.$$

Problem 50. *Compute the following dot products.*

(a) $\langle 1,2,3\rangle \cdot \langle 2,-1,4\rangle$

(b) $\langle 8,-2\rangle \cdot \langle 1,5\rangle$

(c) $(3\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \cdot (2\mathbf{i} - 2\mathbf{j} + \mathbf{k})$

Answers: 12, -2, 1.

For the last one, we are using the answer to Problem 47 to rewrite this as $\langle 3,2,-1\rangle \cdot \langle 2,-2,1\rangle$, since we haven't made any claims about linearity yet.

Problem 51. *Suppose that* $\mathbf{v}$ *and* $\mathbf{w}$ *are vectors. Is* $\mathbf{v}\cdot\mathbf{w}$ *a vector, or a number?*

Possible followup question: So if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are all vectors, can we multiply $\mathbf{u}\cdot\mathbf{v}\cdot\mathbf{w}$? (No.)

Micro-lecture: We have defined this weird multiplication, that for any two vectors gives you a number. Why did we do that? The procedure seems kind of natural, but what is this number measuring? We will answer that question in the next class.

Definition 5. *We say the vectors* $\mathbf{v}$ *and* $\mathbf{w}$ *are* orthogonal *if* $\mathbf{v}\cdot\mathbf{w} = 0$.

Problem 52. *Which of these pairs of vectors are orthogonal?*

(a) $\langle 3,4\rangle$ *and* $\langle 4,-3\rangle$

(b) $\langle -1,-1,4\rangle$ *and* $\langle 2,3,-1\rangle$

(c) $\mathbf{i}+3\mathbf{j}$ *and* $2\mathbf{i}-6\mathbf{j}$

(d) $\mathbf{i}$ *and* $\mathbf{k}$

The fact that orthogonal means perpendicular is a surprise for tomorrow! Answers: (a) and (d) are orthogonal; (b) and (c) are not.

1.6 Day 6

Problem 53. *Find a vector orthogonal to* $\langle 1,3,1\rangle$.

There are many possible correct answers. This problem is reminding them what the dot product was, and what the definition of "orthogonal" was, from yesterday.

Problem 54. *Is it always true that* $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$*? If so, explain why; if not give a counterexample.*

Yes: $u_1v_1+u_2v_2+u_3v_3 = v_1u_1+v_2u_2+v_3u_3$.

Problem 55. *Is it always true that* $\mathbf{u}(\mathbf{w}\cdot\mathbf{v}) = (\mathbf{u}\cdot\mathbf{w})(\mathbf{u}\cdot\mathbf{v})$*? If so, explain why; if not give a counterexample.*

No; in particular the left hand side is a vector and the right side is a scalar. If you had a lot of time, you could ask them to propose alternate formulas that they think *are* true.

Problem 56. *Show that if* $\mathbf{v} = \langle v_1, v_2, v_3\rangle$ *and* $\mathbf{w} = \langle w_1, w_2, w_3\rangle$*, then*

$$|\,\mathbf{v}-\mathbf{w}\,|^2 = |\,\mathbf{v}\,|^2 - 2\,\mathbf{v}\cdot\mathbf{w} + |\,\mathbf{w}\,|^2 .$$

This involves some messy algebra. Also, many students want to start with the whole statement and then simplify until they get $0=0$. If you have time, it's good to talk about why it's better to start with one side and end up with the other.

Problem 57. *Use the Law of Cosines (Theorem 1) and Problem 56 to show that for any two vectors* $\mathbf{v}$ *and* $\mathbf{w}$*,*

$$\mathbf{v}\cdot\mathbf{w} = |\,\mathbf{v}\,|\,|\,\mathbf{w}\,|\cos\theta$$

where θ *is the angle between* $\mathbf{v}$ *and* $\mathbf{w}$*.*

This is a very important property of the dot product! Be sure this gets into the summary at the end of the day.

My students generally do not have a lot of experience with this kind of problem, so they don't always know what I mean by "using" previous results. In particular, they think that if they got stuck in the algebra and couldn't finish Problem 56, that they can't do this problem!

Problem 58. *Use the dot product to find the angle between the vectors* $4\mathbf{i}-2\mathbf{j}$ *and* $2\mathbf{i}+3\mathbf{j}$*. Illustrate with a sketch.*

Answer: About 83 degrees. Students can check their answer by sketching in the plane.

Problem 59. *If two vectors are orthogonal, what is the angle between them? This is also a very important fact!*

Answer: 90 degrees, or $\frac{\pi}{2}$ radians. Unless one of the vectors is the zero vector.

Problem 60. *Find two vectors orthogonal to* $\langle 1,2\rangle$. *How many are there?*

Answer: infinitely many; any scalar multiple of $\langle -2,1\rangle$ will do it. So the set of vectors forms a line perpendicular to the original vector.

Problem 61. *There were many possible answers to Problem 53. Geometrically, what do they represent?*

Each one represents a vector perpendicular to that vector. If you can push students to view them collectively, all based at the same point, they should sweep out a plane perpendicular to the original vector.

Problem 62. *Write an equation for the line through the points* $(5,-3,4)$ *and* $(2,1,-2)$.

Answer: $(x,y,z) = (5,-3,4) + t\langle -3,4,-6\rangle$, or other forms.

Problem 63. *Is the line through* $(-3,-10,5)$ *and* $(-1,-2,3)$ *parallel to the line through* $(4,12,1)$ *and* $(-1,-8,6)$*?*

Yes. The vectors are $\langle 2,8,-2\rangle$ and $\langle -5,-20,5\rangle$.

Problem 64. *(a) Find two points on the plane* $3x-2y+5z=5$ *and one point that is not on the plane.*

(b) Find a vector in (or parallel to) the plane $3x-2y+5z=5$ *and another vector that is not in (or parallel to) the plane.*

(c) Is either of these vectors orthogonal to $\langle 3,-2,5\rangle$*?*

Again, many possible correct answers for part (a). Guess and check is advised. For part (b), use vectors between the points you found in (a). For part (c), yes: the vector in the plane should be perpendicular to $\langle 3,-2,5\rangle$.

This problem is supposed to set up the geometric reasoning for the very important Problem 73 tomorrow. In the discussion, by comparing this problem to Problem , you may be able to get them to conjecture that the plane perpendicular to $\langle a,b,c\rangle$ has an equation of the form $ax+by+cz=d$. But I wouldn't force it.

Definition 6. *If a, b, c, and d are numbers, the* 2×2 determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Problem 65. *Compute the following 2×2 determinants.*

(a) $\begin{vmatrix} 1 & 3 \\ 2 & 2 \end{vmatrix}$

(b) $\begin{vmatrix} 1 & -1 \\ 5 & 3 \end{vmatrix}$

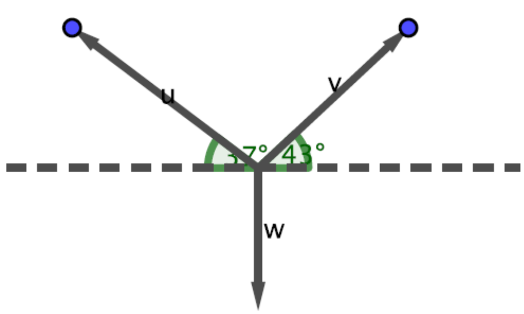
(c) $\begin{vmatrix} 5 & -1 \\ 1 & 2 \end{vmatrix}$

This is to get them ready for the definition of cross product, tomorrow.

Problem 66. *I hung a decorative light on my wall the other day by hanging two hooks on the wall, attaching two ribbons to the light, then tying each ribbon to a hook. It was a little crooked, so one of the ribbons is at an angle of 37 degrees from horizontal and the other is at an angle of 43 degrees from horizontal. The light did not fall down, so the force of tension in the two ribbons must be exactly counteracting the force of gravity on the light (which is about 4 Newtons). Draw a diagram of these three force vectors, and use the fact that their sum is zero to compute the tensions in the two ribbons.*

There is probably not time to discuss this problem on this day. But if one group of students writes their solution on the board, the many students who gave up almost right away might get a hint from seeing what was on the board and try again for the next day.

This is a classic kind of problem that engineering students have to solve to calculate tensions when building a bridge, etc.



Solution: If we let B_1 represent the tension on the left hand ribbon in the diagram, and B_2 represent the tension on the right hand ribbon, then the three force vectors are

$$\mathbf{u} = \langle -B_1 \cos 37^\circ, B_1 \sin 37^\circ \rangle,$$

$$\mathbf{v} = \langle B_2 \cos 43^\circ, B_2 \sin 43^\circ \rangle,$$

$$\mathbf{w} = \langle 0, -4 \rangle.$$

The light isn't moving, so the forces must be exactly canceling each other out: $\mathbf{u}+\mathbf{v}+\mathbf{w}=\mathbf{0}$. This gives us two equations (one for each component) in the two unknowns B_1 and B_2, which we solve to get $B_1 \approx 2.97$ N and $B_2 \approx 3.24$ N.

1.7 Day 7

Problem 67. *Sketch the two vectors* $5\mathbf{i}-\mathbf{j}+\mathbf{k}$ *and* $\mathbf{i}+5\mathbf{j}$*. Are they perpendicular?*

Because of the skewed perspective of the graph, they probably won't look perpendicular, but the dot product tells us they are.

Problem 68. *Use the dot product to find the angles between the vectors in Problems 39 and 41.*

And wasn't that nicer than using the Law of Cosines?

Problem 69. *Is it always true that* $c(\mathbf{v}\cdot\mathbf{w}) = (c\mathbf{v})\cdot\mathbf{w} = \mathbf{v}\cdot(c\mathbf{w})$*? If so, explain why; if not give a counterexample.*

Yes: all are $cv_1w1+cv_2w_2+cv_3w_3$ (in 3D).

Problem 70. *Is it always true that* $\mathbf{u}+(\mathbf{v}\cdot\mathbf{w}) = (\mathbf{u}+\mathbf{v})\cdot(\mathbf{u}+\mathbf{w})$*? If so, explain why; if not give a counterexample.*

No. The left side doesn't even make sense: how do you add a vector to a scalar?

Problem 71. *Is it always true that* $|\mathbf{u}||\mathbf{v}| = |\mathbf{u}\cdot\mathbf{v}|$*? If so, explain why; if not give a counterexample.*

This one looks good, doesn't it? But no. One example might be $\langle 1,2\rangle$ and $\langle 2,1\rangle$. More deeply, compare with Problem 57.

Problem 72. *Is it always true that* $\mathbf{u}\cdot\mathbf{u} = |\mathbf{u}|^2$*? If so, explain why; if not give a counterexample.*

Yes! Both are $u_1^2+u_2^2+u_3^2$. Or you could argue that the angle between $\mathbf{u}$ and itself is 0.

Problem 73. *Suppose that* $P=(p_1,p_2,p_3)$ *and* $Q=(q_1,q_2,q_3)$ *are points in the plane* $3x-2y+5z=7$*.*

(a) What is $3q_1-2q_2+5q_3$*?*

(b) Write a formula for the vector from P *to* Q*:* $\overrightarrow{PQ}=\langle \quad , \quad , \quad \rangle$.

(c) What is $\langle 3,-2,5\rangle\cdot\overrightarrow{PQ}$*?*

(d) What is the angle between $\langle 3,-2,5\rangle$ *and the plane* $3x-2y+5z=7$*? Illustrate with a sketch.*

This problem is very important!!! It sets up the reasoning in the following problem, which is where the equation of the plane comes from.

Answers and notes:

(a) 7. This is what it means for the point to be in the plane. What if we did the same thing with p's? We would also get 7. (We need this fact in part (c)!)

(b) $\langle q_1 - p_1, q_2 - p_2, q_3 - p_3 \rangle$.

(c) Make sure they simplify this and refer back to part (a)! $7 - 7 = 0$. Therefore $\langle 3, -2, 5 \rangle$ and $\overrightarrow{PQ}$ are perpendicular.

(d) The point here is that because we can pick any two points P and Q in the plane, every vector that lies in the plane is perpendicular to $\langle 3, -2, 5 \rangle$. Therefore $\langle 3, -2, 5 \rangle$ is perpendicular to the whole plane. The sketch should show a plane, two points in the plane (labeled P and Q), the vector from P to Q (which lies in the plane), and a vector perpendicular to the plane.

This is a bit backwards from the way Stewart's textbook derives the formula for a plane. Stewart defines a plane geometrically in terms of a point and a normal vector, then proves that the equation will be linear. In these notes we have shown by example and discussed why a linear equation forms a plane (as opposed to a line); we are proving that the vector of coefficients is perpendicular to that plane.

"Slope," for a plane in 3D, can't be just one number. One way to define slope is to give a vector perpendicular to the plane, which is what this problem and the next do (and is the approach taken in Stewart's textbook). Another way is to give two slopes, for example in the xz and yz directions, to form an equation analogous to $(y - y_0) = m(x - x_0)$: $z - z_0 = m_1(x - x_0) + m_2(y - y_0)$. With lots of time, this would be an interesting thing for students to explore, and in fact this idea will come back in Calculus III with partial derivatives and tangent planes.

Problem 74. *(a) Looking back at the previous problem, explain why the vector $\langle a, b, c \rangle$ is always orthogonal to the plane $ax + by + cz = d$.*

(b) Is this consistent with the planes in Problems 6, 13, 24, and 26? Draw pictures to illustrate.

The reasoning is the same as the previous problem, but with a, b, c in place of the numbers. We've sketched all of those planes before, so this should reinforce those ideas.

Problem 75. *(a) Write the equation of a plane perpendicular to the vector $\langle 5, 1, 2 \rangle$. (There should be many possible choices. Why?)*

(b) Which plane, perpendicular to the vector $\langle 5, 1, 2 \rangle$, contains the point $(1, 2, 2)$? Explain your reasoning.

Definition 7. *The* cross product *of the vectors* $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ *and* $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$ *is*

$$\mathbf{v} \times \mathbf{w} = \left\langle \begin{vmatrix} v_2 & v_3 \\ w_2 & w_3 \end{vmatrix}, \begin{vmatrix} v_3 & v_1 \\ w_3 & w_1 \end{vmatrix}, \begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} \right\rangle.$$

Problem 76. *Compute the following cross products.*

(a) $\langle 3, 5, -2 \rangle \times \langle 5, -1, 0 \rangle$

(b) $\mathbf{i} \times \mathbf{j}$

(c) $(-4\mathbf{i} + 3\mathbf{k}) \times (2\mathbf{i} - 5\mathbf{j})$

(d) $\langle 5, -1, 0 \rangle \times \langle 3, 5, -2 \rangle$

Answers: $\langle -2, -10, -28 \rangle$, $\mathbf{k}$, $15\mathbf{i} + 6\mathbf{j} + 20\mathbf{k}$, $\langle 2, 10, 28 \rangle$ (unless I made a mistake).

Talking points: order matters! And, when given vectors in one notation (angle brackets vs **ijk**'s), it is good etiquette to return an answer in the same notation.

Problem 77. *(a) Is the cross product of two vectors a vector, or a scalar?*

(b) Which of these makes sense? $(\mathbf{u} \cdot \mathbf{v}) \times \mathbf{w}$, *or* $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$*?*

1.8 Day 8

Problem 78. *Does the line with parametric equations $x = 2 - t$, $y = -5 + 4t$, and $z = 2 + 3t$ intersect the line with parametric equations $x = 5 + 2t$, $y = -3 - t$, and $z = 9 + 2t$? If so, where? If not, why not?*

Try to get a presentation of this problem from a student who says they don't intersect, because they set the x and y and z equations equal but left the t's the same. Then you don't have to let on that this student is wrong, until after the next problem.

Problem 79. *Which, if either, of the lines in Problem 78 contains the point $(1, -1, 5)$?*

Both lines contain this point. What does that mean for the answer to the previous problem?

Problem 80. *Suppose two toy airplanes are flying along the paths given in Problem 78. Do they collide? What do x, y, z, and t represent in this situation?*

Answers to these three problems: The lines do intersect, at $(1, -1, 5)$, when $t = 1$ in the first line and $t = -2$ in the second line. Both lines contain this point. The two toy airplanes do not collide, because even though they both pass through that point, they are there at different times– maybe 1pm and 10am, respectively.

Problem 81. *Write the equation for the plane which is perpendicular to the line $(2, -1, 6) + t\langle -3, 4, 2\rangle$, and contains the point $(1, 4, -7)$. Sketch the line and the plane.*

Answer: $-3x + 4y + 2z = -1$.

Problem 82. *Write an equation for the line through the point $(1, 2, 4)$ which is perpendicular to the plane $z = -2x - 4y + 12$. Sketch the line and the plane.*

One possible answer: $(1, 2, 4) + t\langle 2, 4, 1\rangle$.

Problem 83. *Prove that the cross product of two vectors is perpendicular to the two original vectors. In other words, $\mathbf{v} \times \mathbf{w}$ is perpendicular to $\mathbf{v}$ and to $\mathbf{w}$.*

Solution: Algebra: compute $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w})$ and $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w})$. Note: this is a very important property!

Problem 84. *Find a vector perpendicular to both $\langle 5, -3, 1\rangle$ and $\langle -4, 2, -1\rangle$.*

Problem 85. *Given two vectors $\mathbf{v}$ and $\mathbf{w}$ in three dimensional space, how many vectors are there that are perpendicular to both $\mathbf{v}$ and $\mathbf{w}$?*

There are infinitely many, but they are all scalar multiples of each other / in the same direction.

Problem 86. *(a) Is it always true that* $\mathbf{v} \times \mathbf{w} = \mathbf{w} \times \mathbf{v}$*? Prove or give a counterexample.*

(b) Is it always true that $(k\mathbf{v}) \times \mathbf{w} = \mathbf{v} \times (k\mathbf{w}) = k(\mathbf{v} \times \mathbf{w})$*? Prove or give a counterexample.*

Answers: (a) No, they are opposites of each other. (b) Yes. (algebra)

Problem 87. *(a) Is it always true that* $(\mathbf{u}+\mathbf{v}) \times \mathbf{w} = \mathbf{u} + (\mathbf{v} \times \mathbf{w})$*? Prove or give a counterexample.*

(b) Is it always true that $\mathbf{u} \times (\mathbf{v}+\mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$*? Prove or give a counterexample.*

Claim 1. *The Right Hand Rule. Suppose you have two vectors* $\mathbf{v}$ *and* $\mathbf{w}$ *in three dimensional space. If you point the fingers of your right hand along* $\mathbf{v}$*, then curl your fingers towards* $\mathbf{w}$ *(rotate your hand so you can do this without hurting yourself!), then your thumb will point in the direction of* $\mathbf{v} \times \mathbf{w}$.

Problem 88. *Sketch the vectors from Problem 76 and check that the Right Hand Rule is true in these cases.*

1.9 Day 9

Problem 89. *Find the equation of the plane containing* $(2,3,4)$, $(1,2,3)$ *and* $(6,-2,5)$.

Use the three points to create two vectors in the plane, then use the cross product to find a vector perpendicular to the plane.

Q: What do you get when you cross a mosquito with a mountain climber?

A: Don't be silly, you can't cross a vector with a scalar.

Problem 90. *Find the equation of the plane containing the line* $(x,y,z) = (1+2t,\ -1+3t,\ 4+t)$ *and the point* $(1,-1,5)$.

Definition 8. *It often happens that we have a vector* $\mathbf{v}$*, and another direction (represented by another vector* $\mathbf{u}$*), and we want to write* $\mathbf{v}$ *as a sum of two vectors, one of which is in the given direction (parallel to* $\mathbf{u}$*) and the other of which is perpendicular to* $\mathbf{u}$*.*

For example, when an object is sliding down an inclined plane, the force of gravity pulls the object directly downwards. This force can be divided into a component parallel to the plane (direction of motion) and another component perpendicular to the plane (which is involved in friction).

We call these two pieces components *of* $\mathbf{v}$*. In particular, one is* the component of $\mathbf{v}$ in the direction of $\mathbf{u}$ *and the other is* the component of $\mathbf{v}$ perpendicular to $\mathbf{u}$.

Problem 91. *Draw a sketch (in the plane!) of the vectors* $\mathbf{v} = -\mathbf{i}+4\mathbf{j}$ *and* $\mathbf{u} = \mathbf{i}+\mathbf{j}$*, and the components of* $\mathbf{v}$ *parallel and perpendicular to* $\mathbf{u}$*.*

Problem 92. *(a) Use your sketch from Problem 91 and some trigonometry to write a formula for the length of the component of* $\mathbf{v}$ *parallel to* $\mathbf{u}$*.*

(b) Now that you know the length, write the component of $\mathbf{v}$ *parallel to* $\mathbf{u}$ *as a scalar multiple of* $\mathbf{u}$*. (Hint: Remember Problem 33?)*

Problem 93. *(a) Compare your formulas in the previous problem with the right hand side of the formula from Problem 57.*

(b) Rewrite your formulas using the dot product.

If the sketch from Problem 91 was careful enough, you can plug the $\mathbf{u}$ and $\mathbf{v}$ from the sketch into the formula and check whether you get something that looks like the component in the sketch.

Problem 94. *In particular, when we calculate work, only the component of the force that is in the direction of the motion matters. Therefore, in the vector version of "Work equals force times distance," when we say "times" we mean the dot product.*

Find the work done by a (constant!) force of 20N directly downward, in sliding a box a distance of 5m along a plane which is at an angle of 45 degrees.

Answer: $50\sqrt{2}$ joules.

Problem 95. *Consider the intersection of the two planes $3x-2y+6z=1$ and $3x-4y+5z=1$.*

(a) What kind of shape will this be? How can we write an equation or equations for that kind of shape? What information do we need to do that? Can we get that information from the two plane equations?

(b) Find the equation(s) for the intersection.

Solution: A vector along the line is in both planes, so it is perpendicular to the normal vectors of the two planes. So we cross the two vectors $\langle 3,-2,6\rangle$ and $\langle 3,-4,5\rangle$ to get $\langle 14,3,-6\rangle$ (or its opposite) as the direction vector for the line. Then we have to find a point that is on the line, i.e. on both planes; guess and check works for this. $(\frac{1}{3},0,0)$ and $(5,1,-2)$ are both possibilities.

Problem 96. *Draw a sketch of two vectors $\mathbf{v}$ and $\mathbf{w}$ in the plane, based at the same point. Imagine the two vectors are two sides of a parallelogram, and draw the other two sides. Use some trigonometry to write a formula for the area of this parallelogram.*

Answer: $|\mathbf{v}||\mathbf{w}|\sin\theta$

Yesterday we discussed the significance of the direction of the cross product. But the cross product is a vector, which has both direction and magnitude. What does the magnitude represent? Here is the answer, in five complicated steps.

Problem 97. *(a) Suppose that $\mathbf{a}=\langle a_1,a_2,a_3\rangle$ and $\mathbf{b}=\langle b_1,b_2,b_3\rangle$. Write out $|\mathbf{a}\times\mathbf{b}|^2$, in gory detail.*

(b) Now write out $|\mathbf{a}|^2|\mathbf{b}|^2-(\mathbf{a}\cdot\mathbf{b})^2$, in gory detail, and compare with your answer in part a.

(c) Explain why $|\mathbf{a}|^2|\mathbf{b}|^2-(\mathbf{a}\cdot\mathbf{b})^2=|\mathbf{a}|^2|\mathbf{b}|^2\sin^2\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. (Hint: Use Problem 57!)

(d) Putting these three problems together, what can we conclude about $|\mathbf{a}\times\mathbf{b}|$?

Parts (a) and (b) should both come out to

$$a_2^2b_3^2+a_3^2b_2^2+a_1^2b_3^2+a_3^2b_1^2+a_1^2b_2^2+a_2^2b_1^2-2a_2a_3b_2b_3-2a_1a_3b_1b_3-2a_1a_2b_1b_2.$$

Part (c) depends on $\sin^2\theta = 1-\cos^2\theta$. In the end we should get $|\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$.

Or, as we discussed in Problem 96, the area of the parallelogram!

Possible question for discussion: We know that in general $\mathbf{a}\times\mathbf{b} \neq \mathbf{b}\times\mathbf{a}$. If we do the cross product in the other order, does the magnitude change? (No.)

Problem 98. *Find the area of the parallelogram with corners at* $(0,0,0)$*,* $(3,-1,5)$*,* $(2,2,1)$*, and* $(5,1,4)$*. (Is this a parallelogram?)*

Solution: $\langle 3,-1,5\rangle \times \langle 2,2,1\rangle = \langle -11,7,8\rangle$, so the area is $\sqrt{121+49+64} = \sqrt{234} \approx 15.3$ square units.

Problem 99. *Find the area of the triangle with corners at* $(0,0,0)$*,* $(3,-1,5)$*, and* $(2,2,1)$*.*

It's half of the parallelogram.

Chapter 2

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